Figure 1.1 Cloud native is an approach to application development aiming at leveraging cloud technologies.

1.1 What is cloud native?

On May 25, 2010, Paul Fremantle, a veteran of the cloud industry, wrote a post on his blog titled "Cloud Native."[1] He was among the first to use the term *cloud native*. At a time when concepts and technologies like microservices, Docker, DevOps, Kubernetes, and Spring Boot didn't yet exist, Fremantle discussed what it takes for "applications and middleware to work well in a cloud environment"—to be *cloud native*—with his team at WSO2.

The key concept explained by Fremantle is that cloud native applications should be specifically designed for the cloud and have properties that take advantage of the cloud environment and the cloud computing model. You can move a traditional application (designed to run *on the ground*) to the cloud, an approach commonly referred to as "lift and shift," but that doesn't make the application *native* to the cloud. Let's see what does.

1.1.1 The Three Ps of Cloud Native

What does it mean for applications to be designed specifically for the cloud? The Cloud Native Computing Foundation (CNCF) answers that question in its cloud native definition:[2]

> *Cloud native technologies empower organizations to build and run scalable applications in modern, dynamic environments such as public, private, and hybrid clouds. Containers, service meshes, microservices, immutable infrastructure, and declarative APIs exemplify this approach.*
>
> *These techniques enable loosely coupled systems that are resilient, manageable, and observable. Combined with robust automation, they allow engineers to make high-impact*